

MCA SEM II (AI)

UNIT 6: AI Applications

Introduction to Natural Language Processing (NLP)

What is NLP?

Natural Language Processing (NLP) is a branch of Artificial Intelligence (AI) that enables computers to understand, interpret, and generate human language. NLP combines computer science, linguistics, and machine learning to facilitate communication between humans and machines.

Objectives of NLP

- Understand human language
- Process text and speech data
- Generate meaningful responses
- Enable human-computer interaction

Applications of NLP

- Chatbots and virtual assistants
- Machine translation
- Speech recognition
- Sentiment analysis
- Text summarization
- Information retrieval

Language Models in NLP

Definition

A language model is a system that predicts the probability of occurrence of words in a sentence. It helps computers understand language patterns and generate meaningful text.

Purpose of Language Models

- Predict next word in a sentence
- Improve speech recognition
- Enable text generation
- Support machine translation

Types of Language Models

1. Statistical Language Models

These models use probability calculations based on word sequences.

Example: N-gram Model

An N-gram predicts a word based on previous words.

Example sentence:

“I am going to the market”

For a bigram model:

- Probability of “market” depends on “the”.

2. Neural Language Models

These models use deep learning techniques such as:

- **Recurrent Neural Networks (RNN):** RNNs are neural networks designed to process sequential data such as text and speech. They remember previous inputs using hidden states, making them useful for language modeling and prediction.
- **Long Short-Term Memory (LSTM):** LSTM is a special type of RNN that can learn long-term dependencies in data. It uses memory cells and gates to avoid the vanishing gradient problem and improve sequence learning.
- **Transformers:** Transformers are deep learning models that use the attention mechanism to process words in parallel. They provide better accuracy and faster training, and are widely used in modern NLP systems like chatbots and translators.

They provide better accuracy and understanding of context.

Example

Input:

“She is drinking ____”

Possible predictions:

- water
- coffee
- juice

Grammars in NLP

Definition

Grammar in NLP refers to a set of rules that define how words combine to form valid sentences.

Grammar helps in:

- Sentence formation
- Syntax checking
- Language understanding

Types of Grammars

1. Regular Grammar

- Simplest type of grammar
- Used in lexical analysis
- Handles simple word patterns

2. Context-Free Grammar (CFG)

Most widely used grammar in NLP.

Example Rule

$S \rightarrow NP + VP$

$NP \rightarrow \text{Noun}$

$VP \rightarrow \text{Verb} + \text{Noun}$

Where:

- S = Sentence
- NP = Noun Phrase
- VP = Verb Phrase

Example Sentence

“Ram eats mango”

Structure:

- NP → Ram
- VP → eats mango

3. Context-Sensitive Grammar

Rules depend on the context of words in the sentence.

Used for more complex language structures.

4. Dependency Grammar

Focuses on relationships between words.

Example:

- “eats” is the main verb
- “Ram” depends on “eats”

Parsing in NLP

Definition

Parsing is the process of analyzing a sentence according to grammar rules to determine its structure and meaning.

It helps identify:

- Subject
- Verb
- Object
- Phrase relationships

Importance of Parsing

- Sentence understanding
- Machine translation
- Question answering systems
- Speech processing

Types of Parsing

1. Top-Down Parsing

- Starts from the root symbol
- Breaks sentence into smaller parts

Example

Sentence:

“The boy plays football”

Parser starts with:

Sentence $\rightarrow$ NP + VP

2. Bottom-Up Parsing

- Starts from input words
- Builds sentence structure step by step

3. Dependency Parsing

Finds dependency relationships between words.

Example:

- Subject $\rightarrow$ boy
- Verb $\rightarrow$ plays
- Object $\rightarrow$ football

4. Constituency Parsing

Divides sentence into phrases.

Example:

[NP The boy] [VP plays football]

Example of Parsing

Sentence:

“The cat drinks milk”

Parse Structure

Sentence

Noun Phrase → The cat

Phrase → drinks milk

NLP Pipeline Overview

1. Text Input: The NLP process begins by taking input in the form of text or speech from the user.

Example: “Book a train ticket.”

2. Tokenization: Tokenization is the process of splitting text into smaller units called tokens such as words or sentences.

Example: “The boy plays cricket” → [The, boy, plays, cricket]

3. Stop-word Removal: Commonly used words that do not add much meaning are removed.

Examples of stop words: “is”, “the”, “and”, “of”.

Example:

“The boy is playing” → “boy playing”

4. Stemming/Lemmatization:

Stemming

Reduces words to their root form by removing suffixes.

Example: “playing” → “play”

Lemmatization

Converts words into their meaningful dictionary form.

Example: “better” → “good”

5. Grammar Analysis: Grammar analysis checks whether the sentence follows grammatical rules and identifies sentence structure.

It helps recognize parts of speech such as nouns, verbs, and adjectives.

Example:

“Ram eats mango”

Ram → Noun

eats → Verb

mango → Noun

6. Parsing : Parsing is the process of analyzing the grammatical structure of a sentence according to grammar rules.

It identifies relationships between words and determines the sentence structure.

Example

Sentence:

“The cat drinks milk.”

Parse Structure

Sentence

Noun Phrase → The cat

Verb Phrase → drinks milk

7.Semantic Analysis: Semantic analysis determines the actual meaning of the sentence after parsing.

It helps the computer understand context and relationships.

Example:

“Apple is sweet” → “Apple” refers to fruit.

8.Response Generation : The final step generates meaningful output or response based on processed information.

Example:

Input: “What is the weather today?”

Output: “Today’s weather is hot.”

Advantages of NLP

- Faster communication
- Automation of language tasks
- Improved customer service
- Better data analysis

Challenges in NLP

- Ambiguity in language
- Different accents and dialects
- Complex grammar structures
- Understanding context

Conclusion

Natural Language Processing enables computers to understand and process human language effectively. Language models help predict words, grammars define sentence structures, and parsing analyzes sentence meaning. Together, they form the foundation of modern AI applications such as chatbots, translators, and voice assistants.